

ENG. OMAR AL-NADHARI

Software Engineer | MSc Computer Science Candidate – Artificial Intelligence

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PROFESSIONAL SUMMARY

Software Engineer and MSc Computer Science candidate specializing in Artificial Intelligence at the University of Pisa, with hands-on experience in backend development, software testing, parallel computing, data analysis, programming education, and IT support. Strong in Python, C++, REST APIs, relational databases, Git, Linux, Docker, and automated testing. Built reproducible software, analytics, and performance-computing projects with CI/CD, benchmarking, and structured technical documentation, alongside professional experience in academic and banking environments.

TECHNICAL SKILLS

Programming Languages: Python, C++, C#, JavaScript, TypeScript, SQL, PHP.

Backend & Web: FastAPI, REST APIs, Node.js, Laravel, React.js, HTML/CSS, API development and integration.

Databases: MySQL, SQLite, relational database design, joins, aggregations, subqueries, CTEs, window functions.

Software Engineering & Testing: Object-Oriented Programming, data structures and algorithms, debugging, code review, software testing, pytest, requirements analysis, technical documentation.

DevOps & Tools: Git, GitHub, Docker, GitHub Actions, CI/CD, Linux, Bash, GCC, Makefiles.

Systems & Parallel Computing: OpenMP, MPI, AVX2 SIMD, multithreading, parallel algorithms, SLURM, performance benchmarking, speedup and scalability analysis.

AI & Data: Pandas, NumPy, scikit-learn, SciPy, statsmodels, Jupyter Notebook, data cleaning, exploratory data analysis, feature engineering, classification, cross-validation, class-imbalance handling, model evaluation.

EXPERIENCE

Lecturer - Programming and Software Engineering

Faculty of Engineering, Taiz University, Taiz, Yemen | August 2024 – May 2025

- Entrusted with full responsibility for planning and delivering an undergraduate programming course, including theoretical lectures, practical laboratory sessions, assignments, assessments, and student support.
- Designed programming exercises and software projects and reviewed student code for logical, syntactic, runtime, implementation, and design errors.
- Evaluated student code, software projects, technical documentation, use cases, and system-design diagrams for correctness, completeness, consistency, and adherence to requirements.
- Provided detailed, evidence-based written and verbal feedback while maintaining consistent assessment standards across student submissions.
- Authored a complete course book containing structured explanations, worked examples, and programming exercises.

Research Mentor

Taiz University, Taiz, Yemen | August 2023 – Present

- Guide undergraduate students in research design, methodology, data analysis, interpretation of results, and technical and academic documentation.
- Review research and graduation-project artifacts, including requirements, use cases, software-analysis diagrams, methodologies, and analytical conclusions for correctness, consistency, and completeness.

IT Support and Data Analysis Assistant

AlKuraimi Islamic Bank, Taiz, Yemen | January 2025 – May 2025

- Provided technical support to bank employees, diagnosing and resolving operating-system, application, user-account, permission, printer, and basic network issues.
- Installed and configured Windows systems, Microsoft applications, and banking software while coordinating incidents with central IT and networking teams.
- Collected, cleaned, validated, and organized operational data used in internal analysis and reporting, identifying recurring issues and patterns.
- Supported business requirements analysis for an internal banking workflow, including process flowcharts and functional and non-functional requirements.

SELECTED SOFTWARE & TECHNICAL PROJECTS

Developer Workspace Assistant — *Python, FastAPI, REST APIs, MCP, SQLite, Docker, Pytest*

- Built a FastAPI backend service for managing developer tasks through REST and MCP interfaces, supporting create, retrieve, update, and delete operations.
- Implemented persistent SQLite storage and structured request/response validation for reliable API behavior.
- Containerized the application with Docker and added automated Pytest coverage, with 8 tests passing for core backend functionality.

Parallel Partitioned Hash Join - *C++20, OpenMP, MPI, Multithreading, Linux, GCC, Makefiles, SLURM*

- Developed sequential, multithreaded, OpenMP, and distributed MPI implementations of a partitioned hash-join algorithm.
- Designed and executed controlled performance experiments across different thread and process counts, measuring execution time, throughput, speedup, scalability, workload distribution, and communication or synchronization overhead.
- Evaluated correctness and reproducibility by comparing join counts and checksums across implementations and documented the methodology, commands, results, and limitations.

Bank Marketing Analytics & Subscription Prediction - *Python, Pandas, NumPy, SQL, SQLite, SciPy, statsmodels, scikit-learn, Jupyter Notebook, pytest*

- Built an end-to-end machine-learning workflow on the UCI Bank Marketing dataset with 45,211 customer records, covering validation, cleaning, SQL analysis, feature engineering, and modeling.
- Developed a leakage-safe pre-campaign prediction pipeline by excluding information unavailable at prediction time and using stratified cross-validation.
- Compared baseline and class-balanced models and tuned a Balanced Random Forest that achieved 0.738 ROC-AUC and 0.390 PR-AUC on the held-out test set.
- Selected the decision threshold from training-only out-of-fold predictions, achieving approximately 40% precision, 40% recall, and 3.42× lift; implemented 33 automated Pytest tests.

EDUCATION

MSc in Computer Science – Artificial Intelligence Curriculum

University of Pisa, Italy | September 2025 – Present

Bachelor of Engineering in Information Technology Engineering

Taiz University, Yemen | August 2018 – June 2023

LANGUAGES & HONORS

Languages: Arabic – Native | English – B2

Honor: Tuition Fee Waiver Scholarship, Taiz University – awarded for three consecutive academic years, (2020/2021–2022/2023).